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SECTION: 11 - LINNAEUS

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JAVASCRIPT CONDITIONAL STATEMENTS WITH FORM VALIDATION

Subject: ICT / Programming
Grade Level: Grade 11
Time Allotment: 1.5 Hours

I. Learning Objectives

- Identify and use if, if-else, and if-else-if statements
- Create HTML forms to collect user input
- Apply JavaScript form validation
- Display appropriate output based on conditions

II. Materials Needed

- Computer or laptop
- Text editor (VS Code / Notepad)
- Web browser

III. General Instructions

Follow each step carefully. Type the codes exactly as shown. Save your files frequently and test them using a web browser.

IV. Activity 1: if Statement – Voting Eligibility Checker

Objective: Use an if statement with form validation to determine voting eligibility.

Step 1: HTML Form

Create a file named if.html and type the following code:

```
<!DOCTYPE html>
<html>
<head>
  <title>Voting Eligibility</title>
</head>
<body>
<h3>Voting Eligibility Checker</h3>

<form>
  Enter your age:
  <input type="number" id="age">
  <button type="button" onclick="checkAge()">Check</button>
</form>

<p id="result"></p>

<script src="if.js"></script>
```

```
</body>
</html>
```

Step 2: JavaScript with Validation

Create a file named if.js and type:

```
function checkAge() {
  let age = document.getElementById("age").value;

  if (age === "") {
    document.getElementById("result").innerHTML =
    "Please enter your age.";
  } else if (age >= 18) {
    document.getElementById("result").innerHTML =
    "You are eligible to vote.";
  } else {
    document.getElementById("result").innerHTML =
    "You are not eligible to vote.";
  }
}
```

Guide Question: What happens if the input field is empty?

If the input field is left empty, the JavaScript function will trigger the first if condition. Specially, the variable age will be an empty string. Since the condition if age if (age === "") evaluates true, the program will update the HTML

V. Activity 2: if-else Statement – Password Verification

Objective: Use if-else statement to validate password input.

HTML Code

```
<form>
  Enter Password:
  <input type="password" id="password">
  <button type="button" onclick="checkPassword()">Login</button>
</form>

<p id="message"></p>
```

JavaScript Code

```
function checkPassword() {
  let password =
```

```
  document.getElementById("password").value;

  if (password === "") {
    document.getElementById("message").innerHTML =
    "Password cannot be empty.";
  } else if (password === "admin123") {
    document.getElementById("message").innerHTML =
    "Access Granted";
  } else {
    document.getElementById("message").innerHTML =
    "Access Denied";
  }
}
```

VI. Activity 3: if-else-if Statement – Grade Evaluation

Objective: Use if-else-if statement with range validation.

```
function evaluateGrade() {
  let grade =
  document.getElementById("grade").value;

  if (grade === "") {
    output.innerHTML = "Please enter a grade.";
  } else if (grade < 0 || grade > 100) {
    output.innerHTML =
    "Grade must be between 0 and 100.";
  } else if (grade >= 90) {
    output.innerHTML = "Outstanding";
  } else if (grade >= 80) {
    output.innerHTML = "Very Satisfactory";
  } else if (grade >= 75) {
    output.innerHTML = "Satisfactory";
  } else {
    output.innerHTML =
    "Did Not Meet Expectations";
  }
}
```

VII. Assessment Questions

1. What is the purpose of form validation?
   The purpose of form validation is to ensure that data entered by a user is accurate, complete and in the correct format before it is processed by the program.
2. When should if-else-if be used?
   if-else-if ladder should be used when you have multiple mutually exclusive conditions to check
3. Why is input checking important in programming?
   Input checking is used For Preventing Errors, Security, and User Experience.

VIII. Reflection

What did you learn about conditional statements and validation?

Through this exercise, I learned that conditional statements are the 'brain' of a program, allowing it to make decisions based on specific criteria. I discovered that an if statement handles a single choice, but if-else-if is essential for handling multiple ranges, such as grading scales or multi-step eligibility. Regarding validation, I learned that a programmer cannot assume a user will always followinstructions. By implementing checks for empty strings ("") or out-of-bounds numbers, I can make my code more 'robust.' This means the program won't just fail, it will communicate with the user and guide them toward the correct input, which is a fundamental part of creating professional software."